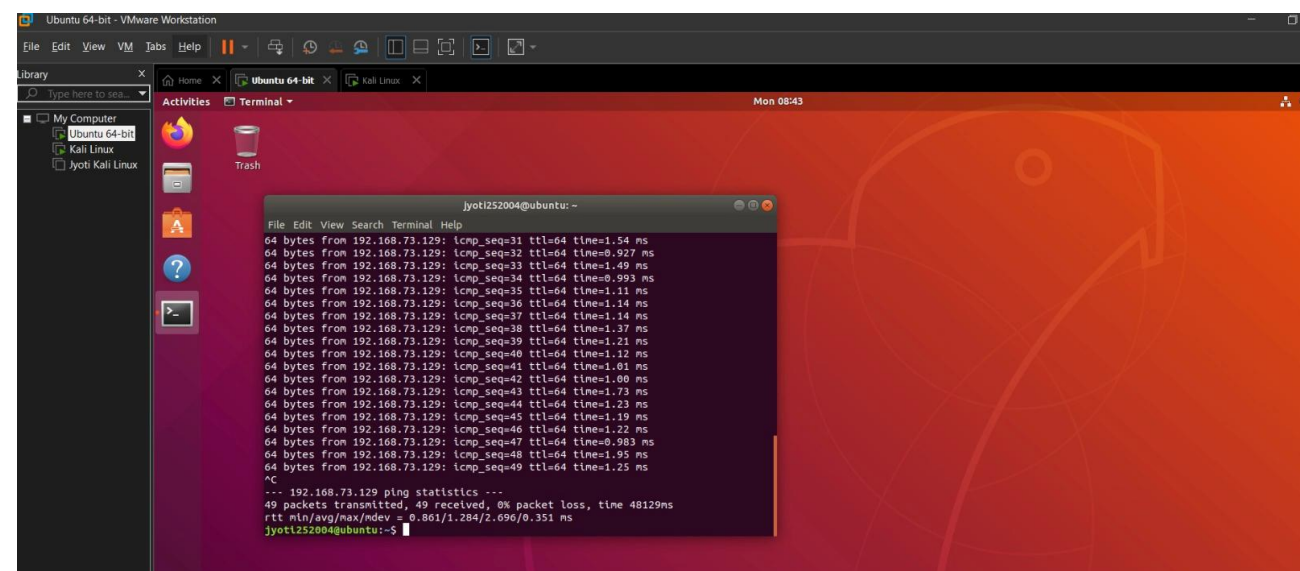
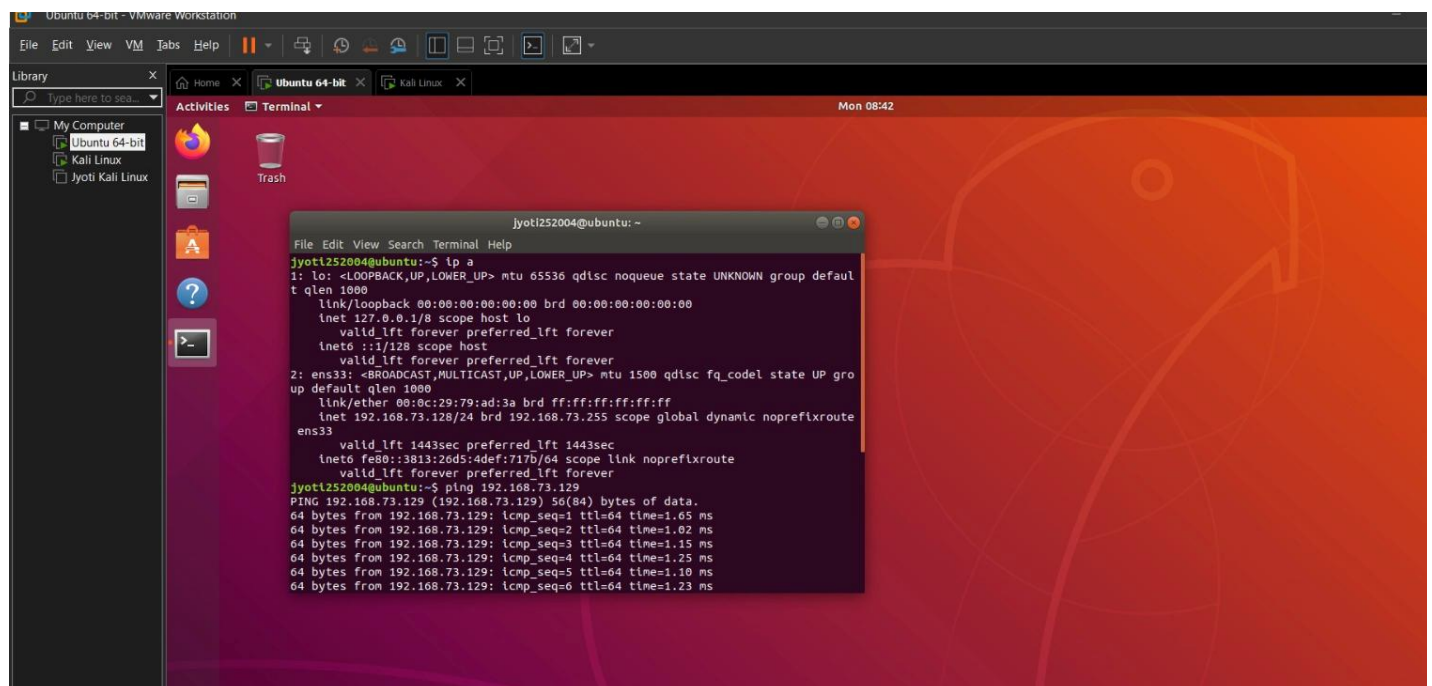
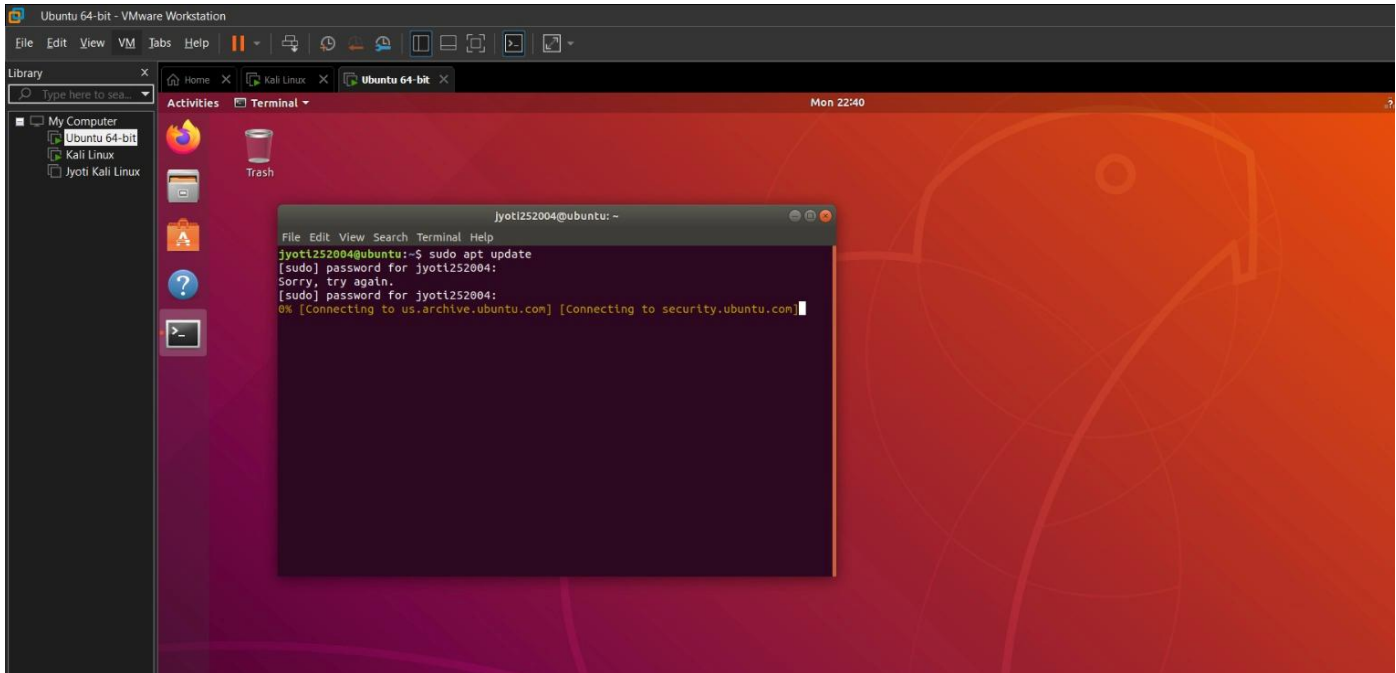


Ubuntu Setup



```
Activities Terminal
Tue 00:13
jyoti252004@ubuntu: ~
File Edit View Search Terminal Help
jyoti252004@ubuntu:~$ sudo apt update
[sudo] password for jyoti252004:
Get:1 http://security.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Hit:2 http://us.archive.ubuntu.com/ubuntu bionic InRelease
Get:3 http://us.archive.ubuntu.com/ubuntu bionic-updates InRelease [102 kB]
Get:4 http://security.ubuntu.com/ubuntu bionic-security/main amd64 DEP-11 Metadata [76.9 kB]
Get:5 http://security.ubuntu.com/ubuntu bionic-security/restricted amd64 DEP-11 Metadata [212 B]
Get:6 http://security.ubuntu.com/ubuntu bionic-security/universe amd64 DEP-11 Metadata [61.8 kB]
Get:7 http://security.ubuntu.com/ubuntu bionic-security/multiverse amd64 DEP-11 Metadata [2,468 B]
Get:8 http://us.archive.ubuntu.com/ubuntu bionic-backports InRelease [102 kB]
Get:9 http://us.archive.ubuntu.com/ubuntu bionic-updates/main amd64 DEP-11 Metadata [297 kB]
Get:10 http://us.archive.ubuntu.com/ubuntu bionic-updates/restricted amd64 DEP-11 Metadata [212 B]
Get:11 http://us.archive.ubuntu.com/ubuntu bionic-updates/universe amd64 DEP-11 Metadata [303 kB]
Get:12 http://us.archive.ubuntu.com/ubuntu bionic-updates/multiverse amd64 DEP-11 Metadata [2,468 B]
Get:13 http://us.archive.ubuntu.com/ubuntu bionic-backports/main amd64 DEP-11 Metadata [8,124 B]
Get:14 http://us.archive.ubuntu.com/ubuntu bionic-backports/restricted amd64 DEP-11 Metadata [216 B]
Get:15 http://us.archive.ubuntu.com/ubuntu bionic-backports/universe amd64 DEP-11 Metadata [10.0 kB]
Get:16 http://us.archive.ubuntu.com/ubuntu bionic-backports/multiverse amd64 DEP-11 Metadata [216 B]
Fetched 1,068 kB in 6s (176 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
jyoti252004@ubuntu:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  fonts-liberation gstreamer1.0-gtk3 libboost-date-time1.65.1 libboost-filesystem1.65.1 libboost-iostreams1.65.1 libboost-locale1.65.1 libbcrdr-0.1-1 libclucene-contribs1v5 libclucene-core1v5
  libcms-0.5v5 libcolamd2 libdazl-1.0-0 libe-book-0.1-1 libedataserver1.2-2 libeot0 libepubgen-0.1-1 libetonyek-0.1-1 libevent-2.1-6 libexiv2-14 libfreerdp-client2-2
  libfreerdp2-2 libgic2 libgee-0.8-2 libgexiv2-2 libgion-1.0-0 libgpgmepp0 libgpgod4 liblangtag-common liblangtag1 liblirc-client0 liblua5.3-0 libmediaart-2.0-0
  libmpeg2-0-1 libmpeg2v5 libraw16 librevenge-0.0-0 libsgutil2-2 libssh4 libsysparseconf5 libvncclient1 libwinpr2-2 libxaplan30 libx11sec1-ss lp-solve
  media-player-info python3-mako python3-markupsafe syslinux syslinux-common syslinux-legacy usb-creator-common
Use 'sudo apt autoremove' to remove them.
The following security updates require Ubuntu Pro with 'esm-infra' enabled:
  libpamg libsoup-gnome4-4-1 bluez libcroco3 libwebp6 open-vn-tools-desktop
  libkrb5-3 libkspapi-krb5-2 libpython3.6-minimal poppler-utils libnghttp2-14
  gstreamer1.0-alsa libiscfg160 libcupss2 intel-microcode update-manager-core
  gnupg-utils xserver-common vin-common gir1.2-soup-2.4 libapt-inst2.0
  gpg-wks-client gstreamer1.0-plugins-base-apps libldap-2.4-2 libpam-modules
  openssl liblootdev-swap2 blues-cups libdri1 gir1.2-gdkpixbuf-2.0 gnupg-lib0n
  libgdk-pixbuf2-0 libc6-dbg libssh4 imagemagick libwebclient0
```

Activities Firefox Web Browser

Tue 00:18

jyoti252004@ubuntu: ~

File Edit View Search Terminal Help

55220 /var/ossec/bin/wazuh-authd
55226 /var/ossec/bin/wazuh-db
55259 /var/ossec/bin/wazuh-apid
55263 /var/ossec/framework
55266 /var/ossec/framework
55269 /var/ossec/framework
55285 /var/ossec/bin/wazuh-apid
55327 /var/ossec/bin/wazuh-apid
55343 /var/ossec/bin/wazuh-apid
55371 /var/ossec/bin/wazuh-apid
55390 /var/ossec/bin/wazuh-apid
55401 /var/ossec/bin/wazuh-apid

Feb 16 23:22:00 ubuntu env[55124]: Star
Feb 16 23:22:01 ubuntu env[55124]: Star
Feb 16 23:22:02 ubuntu env[55124]: Star
Feb 16 23:22:03 ubuntu env[55124]: Star
Feb 16 23:22:03 ubuntu env[55124]: Star
Feb 16 23:22:03 ubuntu env[55124]: Star
Feb 16 23:22:04 ubuntu env[55124]: Star
Feb 16 23:22:04 ubuntu env[55124]: Star
Feb 16 23:22:06 ubuntu env[55124]: Comp
Feb 16 23:22:06 ubuntu systemd[1]: Star
jyoti252004@ubuntu:~\$ sudo systemctl is
active
jyoti252004@ubuntu:~\$ sudo systemctl is
active
jyoti252004@ubuntu:~\$ sudo systemctl is
active
jyoti252004@ubuntu:~\$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536
link/loopback 00:00:00:00:00:00 brd
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft
inet6 ::1/128 scope host
valid_lft forever preferred_lft
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP>
link/ether 00:0c:29:79:ad:3a brd ff:ff:ff:ff:ff:ff
inet 192.168.73.130/24 brd 192.168.73.255
valid_lft 1268sec preferred_lft
inet6 fe80::3813:26d5:4def:717b/64
valid_lft forever preferred_lft
jyoti252004@ubuntu:~\$ AC
jyoti252004@ubuntu:~\$

Welcome to Firefox

Firefox Privacy Notice

https://

Sign in

Open up an amazing internet

Our latest version is built around you, making it easier than ever to zip around the web. It's packed with features we think you'll adore.

Set up in seconds

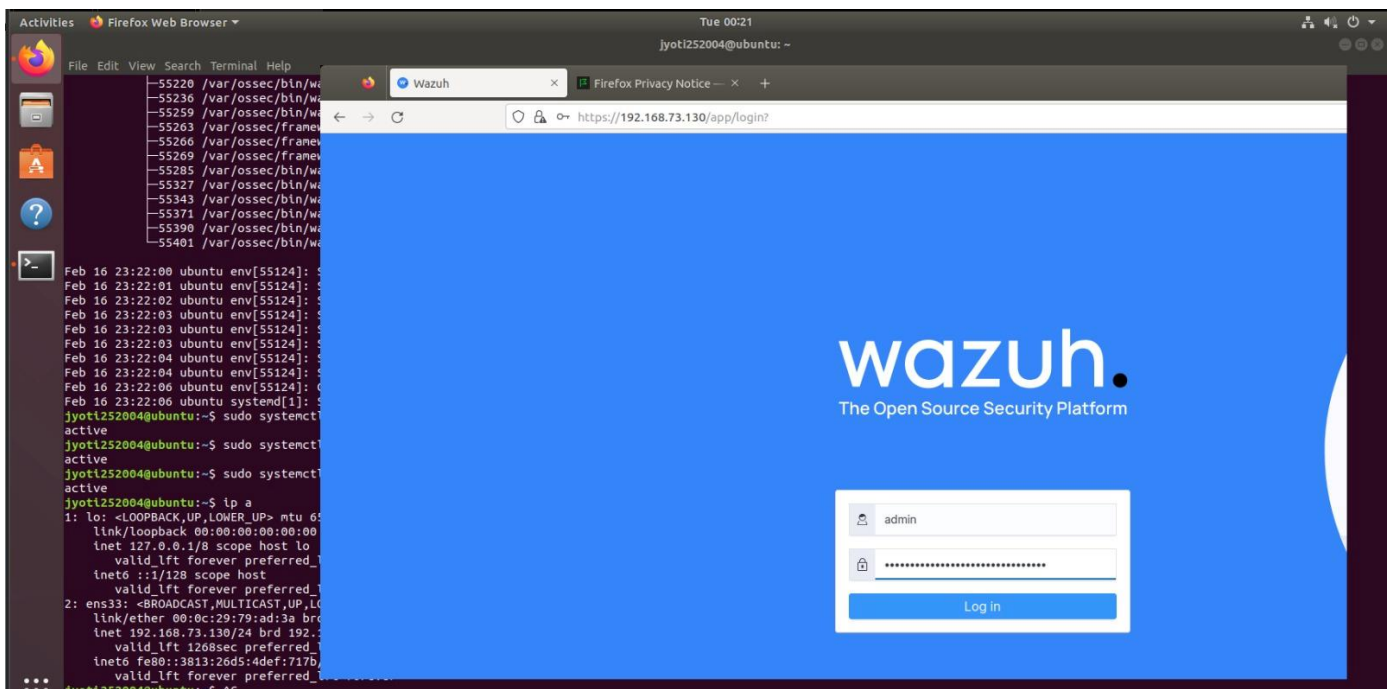
Skip this step

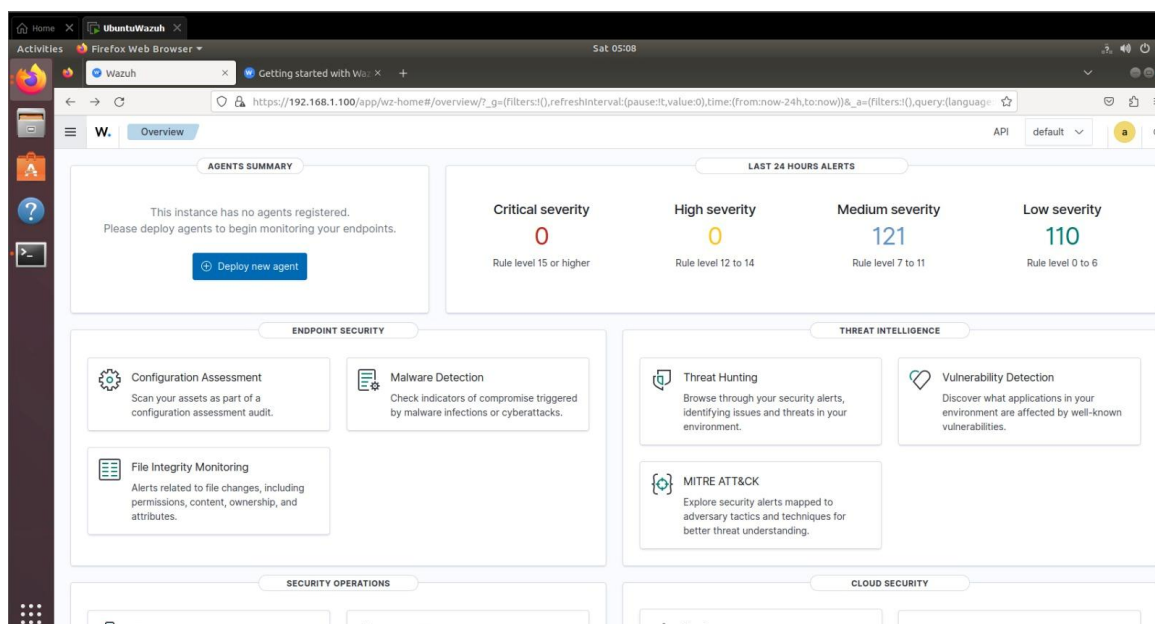
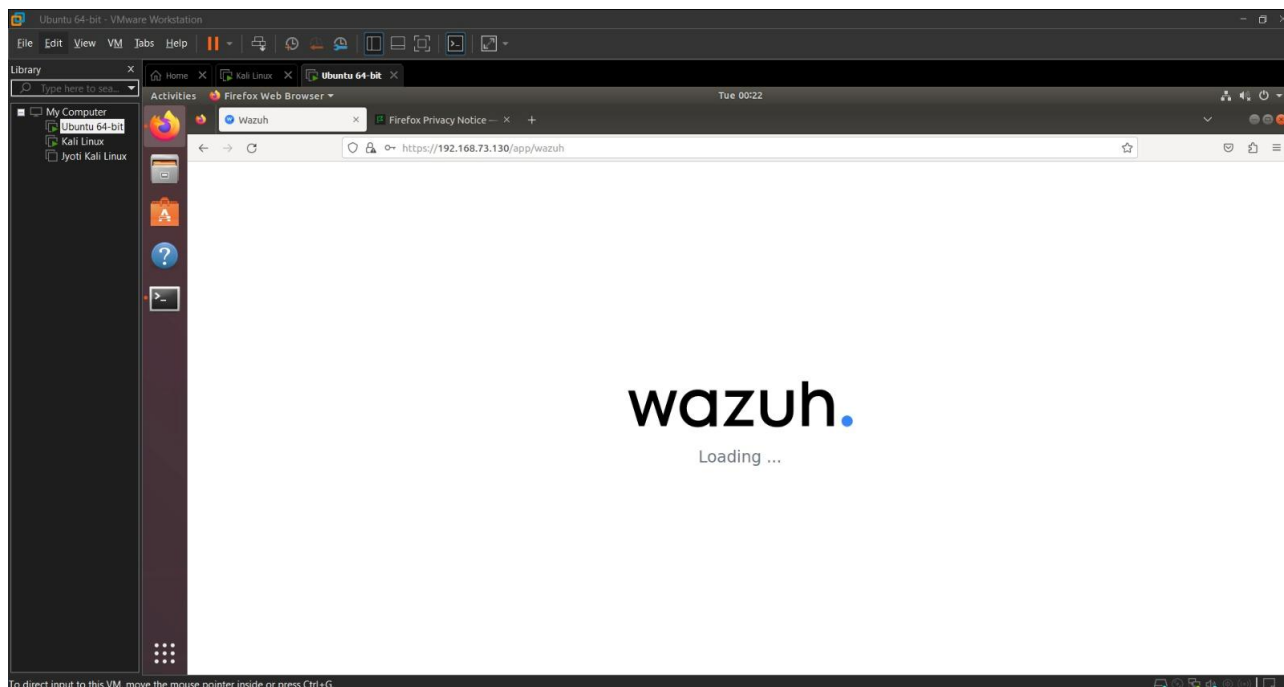
```
Activities Terminal
Tue 00:13
jyoti252004@ubuntu: ~
File Edit View Search Terminal Help
Setting up libcurl4:amd64 (7.58.0-2ubuntu3.24) ...
Setting up curl (7.58.0-2ubuntu3.24) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
Processing triggers for libc-bin (2.27-2ubuntu1.0) ...
jyoti252004@ubuntu:~$ curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
# /bin/bash
# Wazuh Installer
# Copyright (C) 2015, Wazuh Inc.
#
# This program is a free software; you can redistribute it
# and/or modify it under the terms of the GNU General Public
# License (version 2) as published by the FSF - Free Software
# Foundation.
readonly repoppg="https://packages.wazuh.com/key/GPG-KEY-WAZUH"
readonly repobaseurl="https://packages.wazuh.com/4.x"
readonly reporelease="stable"
readonly filebeat_wazuh_module="{repobaseurl}/filebeat/wazuh-filebeat-0.3.tar.gz"
readonly bucket="packages.wazuh.com"
readonly repository="4.x"
adminpem="/etc/wazuh-indexer/certs/admin.pem"
adminkey="/etc/wazuh-indexer/certs/admin-key.pem"
readonly wazuh_major="4.7"
readonly wazuh_version="4.7.5"
readonly filebeat_version="7.10.2"
readonly wazuh_install_veston="0.1"
readonly source_branch="v5{wazuh_version}"
readonly resources="https://{bucket}/{wazuh_major}"
readonly base_url="https://{bucket}/{repository}"
base_path="{dirname "${readlink -f "$0"}")"
readonly base_path
config_file="{base_path}/config.yml"
readonly tar_file_name="wazuh-install-files.tar"
tar_files="{base_path}/{tar_file_name}"
readonly filebeat_wazuh_template="https://raw.githubusercontent.com/wazuh/wazuh/{source_branch}/extensions/elasticsearch/7.x/wazuh-template.json"
readonly dashboard_cert_path="/etc/wazuh-dashboard/certs"
readonly filebeat_cert_path="/etc/filebeat/certs"
readonly indexer_cert_path="/etc/wazuh-indexer/certs"
readonly logfile="/var/log/wazuh-install.log"
debug=> {logfile} 2>&1
readonly base_dest_folder="wazuh-offline"
readonly manager_deb_base_url="{base_url}/apt/pool/main/w/wazuh-manager"
```


Wazuh Setup

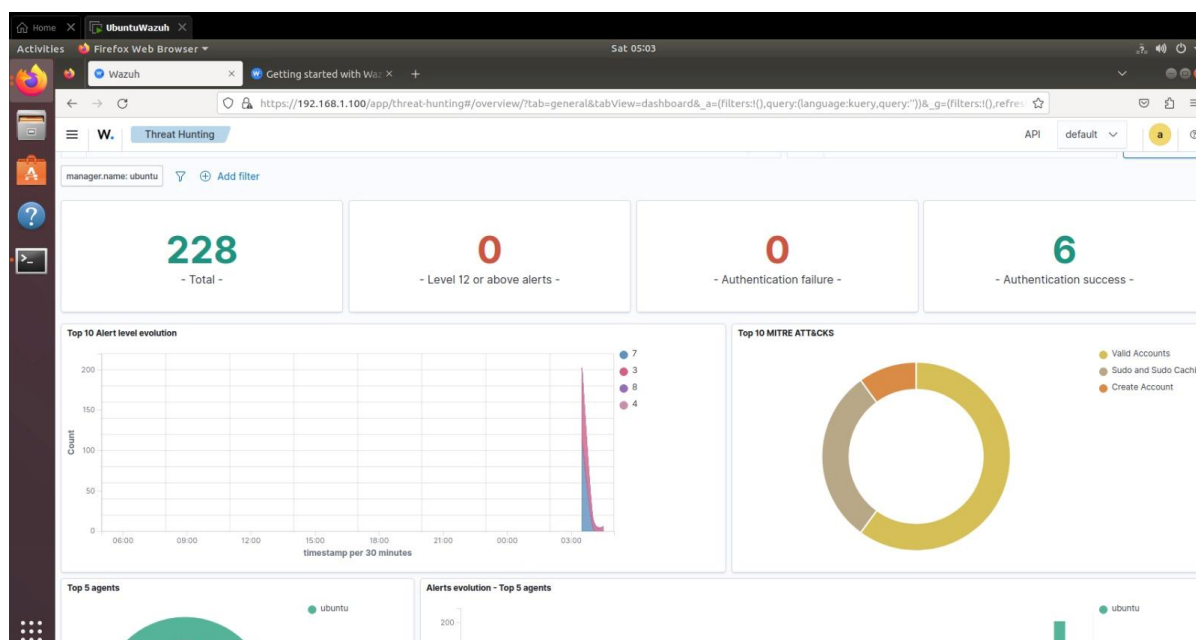
```
18/02/2026 03:20:20 INFO: Starting service wazuh-manager.
18/02/2026 03:20:40 INFO: wazuh-manager service started.
18/02/2026 03:20:40 INFO: Starting Filebeat installation.
18/02/2026 03:20:58 INFO: Filebeat installation finished.
18/02/2026 03:20:59 INFO: Filebeat post-install configuration finished.
18/02/2026 03:20:59 INFO: Starting service filebeat.
18/02/2026 03:21:00 INFO: filebeat service started.
18/02/2026 03:21:00 INFO: --- Wazuh dashboard ---
18/02/2026 03:21:00 INFO: Starting Wazuh dashboard installation.
18/02/2026 03:22:48 INFO: Wazuh dashboard installation finished.
18/02/2026 03:22:48 INFO: Wazuh dashboard post-install configuration finished.
18/02/2026 03:22:48 INFO: Starting service wazuh-dashboard.
18/02/2026 03:22:49 INFO: wazuh-dashboard service started.
18/02/2026 03:23:19 INFO: Initializing Wazuh dashboard web application.
18/02/2026 03:23:19 INFO: Wazuh dashboard web application not yet initialized. Waiting...
18/02/2026 03:23:36 INFO: Wazuh dashboard web application not yet initialized. Waiting...
18/02/2026 03:23:51 INFO: Wazuh dashboard web application initialized.
18/02/2026 03:23:51 INFO: --- Summary ---
18/02/2026 03:23:51 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: F6ewTU619bZoYpa6c*Nycp0JM*qe2GpD
18/02/2026 03:23:51 INFO: --- Dependencies ---
18/02/2026 03:23:51 INFO: Removing gawk.
18/02/2026 03:24:02 INFO: Installation finished.
mtu@ubuntu:~$
```

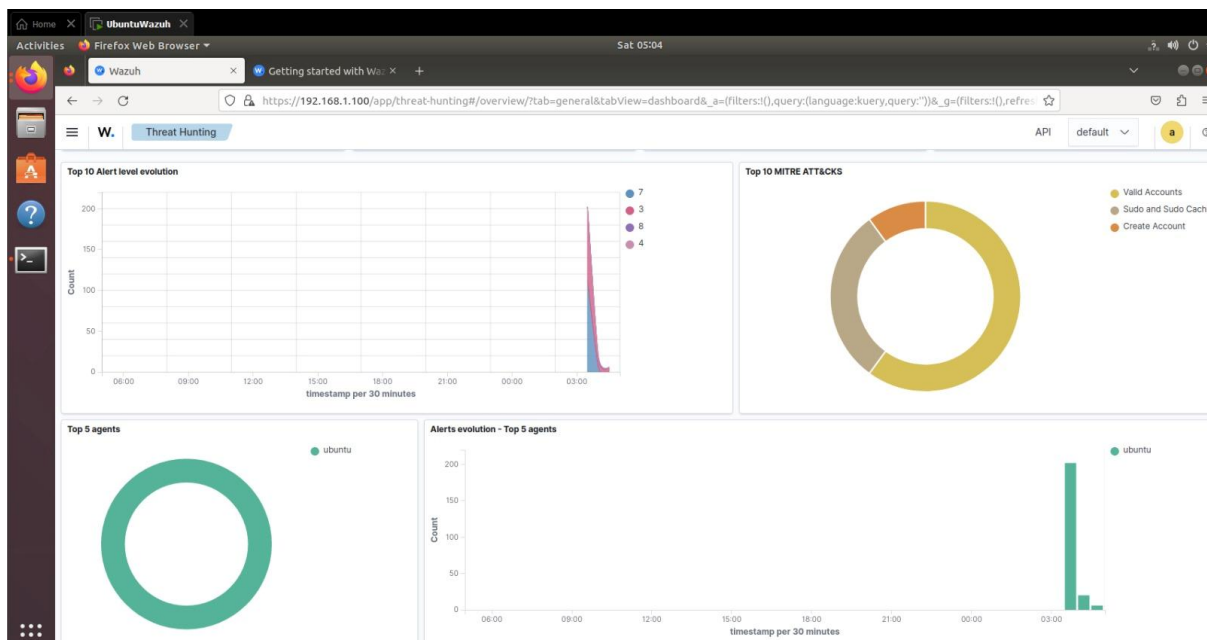
Wazuh Dashboard





Threat Detection Alerts





UbuntuWazuh Threat Hunting

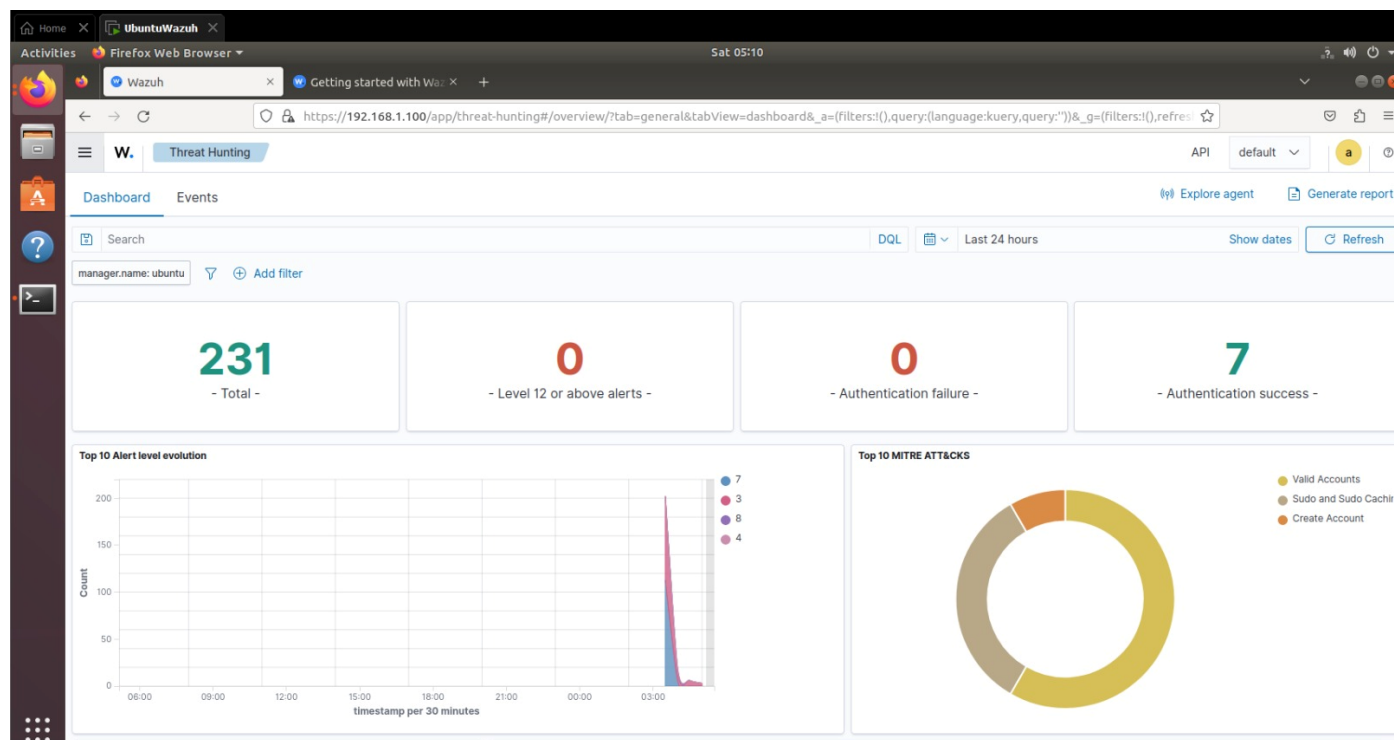
Dashboard Events

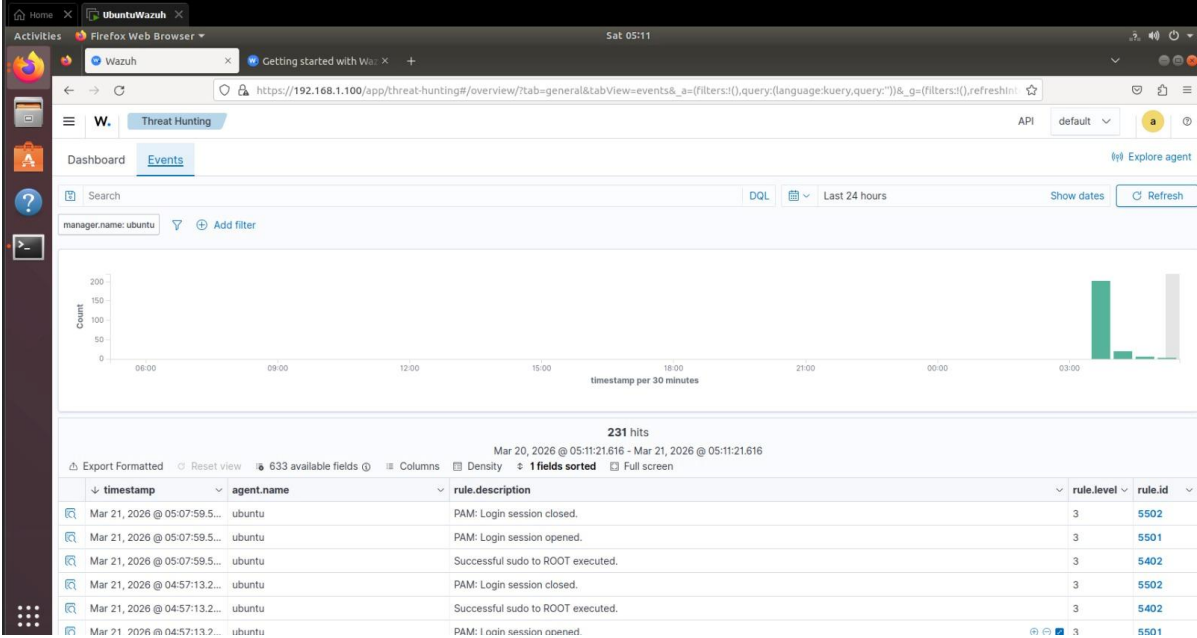
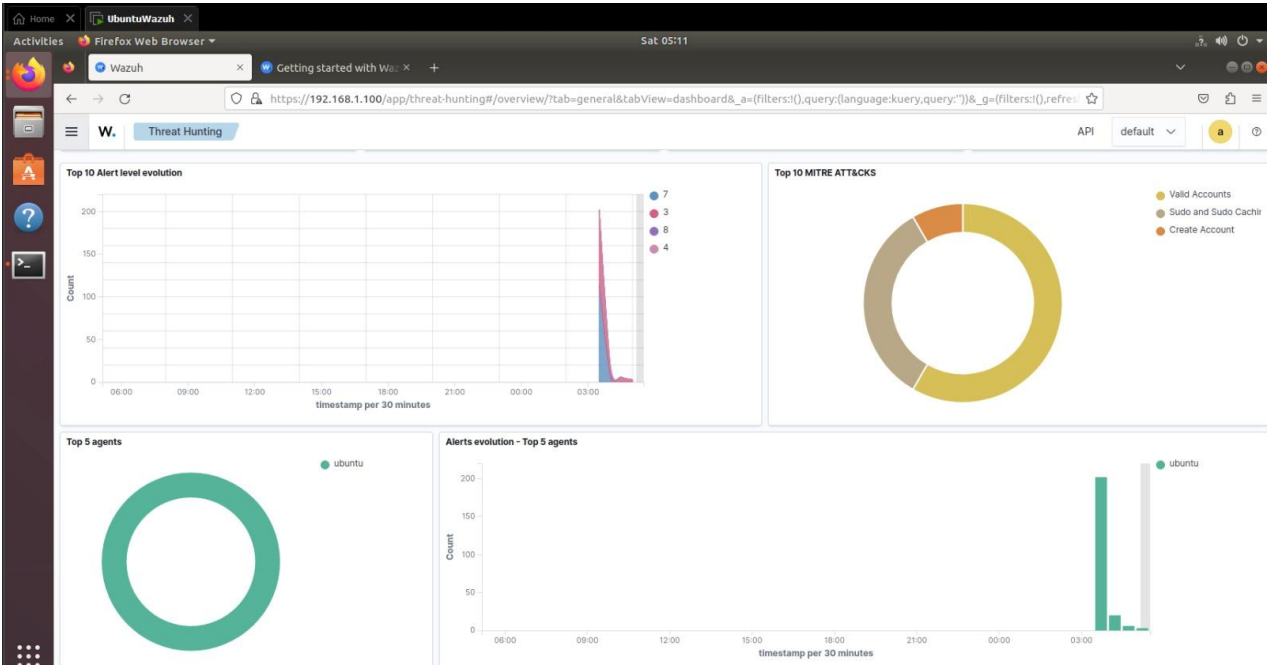
Search: manager.name: ubuntu

228 hits

Mar 20, 2026 @ 05:05:18.532 - Mar 21, 2026 @ 05:05:18.533

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 21, 2026 @ 04:57:13.2...	ubuntu	Successful sudo to ROOT executed.	3	5402
Mar 21, 2026 @ 04:57:13.2...	ubuntu	PAM: Login session opened.	3	5501
Mar 21, 2026 @ 04:57:13.2...	ubuntu	PAM: Login session closed.	3	5502
Mar 21, 2026 @ 04:33:12.4...	ubuntu	Successful sudo to ROOT executed.	3	5402
Mar 21, 2026 @ 04:33:12.4...	ubuntu	PAM: Login session opened.	3	5501





Wazuh Threat Hunting Events

Search: manager.name: ubuntu

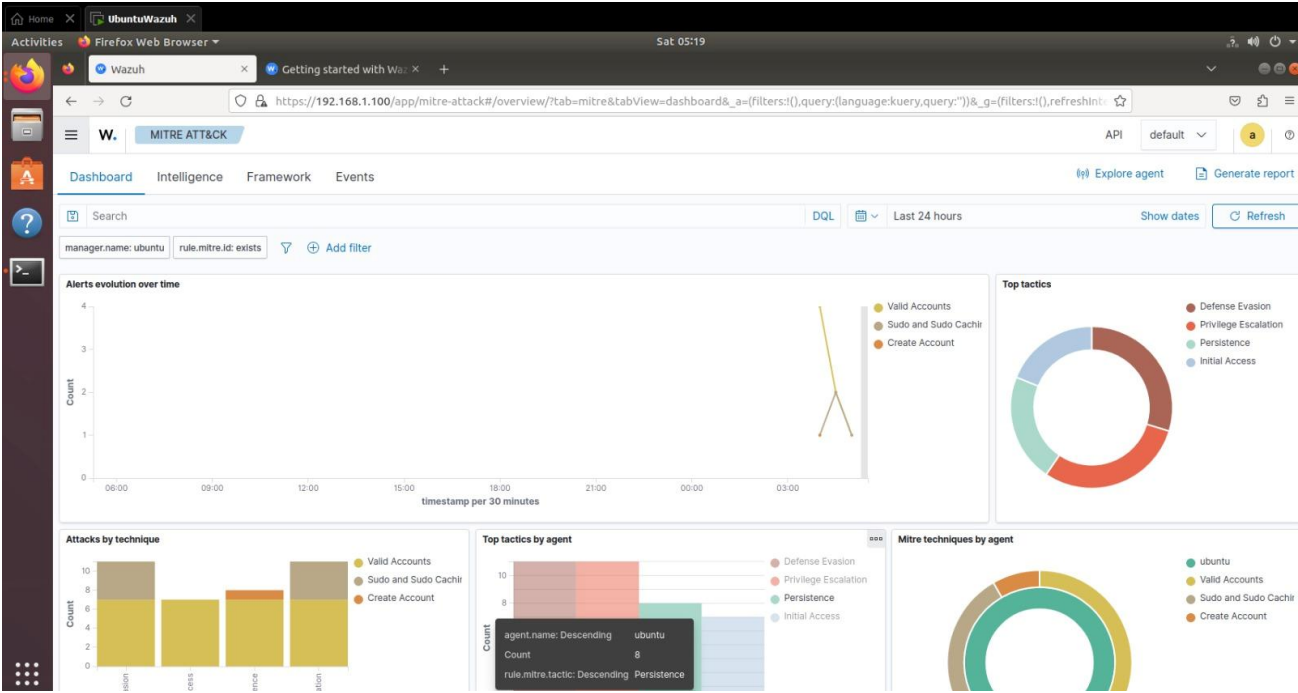
231 hits

Mar 20, 2026 @ 05:24:47.538 - Mar 21, 2026 @ 05:24:47.538

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 21, 2026 @ 05:07:59.5...	ubuntu	PAM: Login session closed.	3	5502
Mar 21, 2026 @ 05:07:59.5...	ubuntu	PAM: Login session opened.	3	5501
Mar 21, 2026 @ 05:07:59.5...	ubuntu	Successful sudo to ROOT executed.	3	5402
Mar 21, 2026 @ 04:57:13.2...	ubuntu	PAM: Login session closed.	3	5502
Mar 21, 2026 @ 04:57:13.2...	ubuntu	Successful sudo to ROOT executed.	3	5402
Mar 21, 2026 @ 04:57:13.2...	ubuntu	PAM: Login session opened.	3	5501
Mar 21, 2026 @ 04:33:12.4...	ubuntu	PAM: Login session opened.	3	5501
Mar 21, 2026 @ 04:33:12.4...	ubuntu	PAM: Login session closed.	3	5502
Mar 21, 2026 @ 04:33:12.4...	ubuntu	Successful sudo to ROOT executed.	3	5402
Mar 21, 2026 @ 04:28:42.2...	ubuntu	PAM: Login session closed.	3	5502
Mar 21, 2026 @ 04:28:42.2...	ubuntu	First time user executed sudo.	4	5403
Mar 21, 2026 @ 04:28:42.2...	ubuntu	PAM: Login session opened.	3	5501
Mar 21, 2026 @ 04:20:16.1...	ubuntu	Listened ports status (netstat) changed (new port opened or closed).	7	533
Mar 21, 2026 @ 04:14:16.7...	ubuntu	Listened ports status (netstat) changed (new port opened or closed).	7	533
Mar 21, 2026 @ 04:08:16.1...	ubuntu	Listened ports status (netstat) changed (new port opened or closed).	7	533

Rows per page: 15

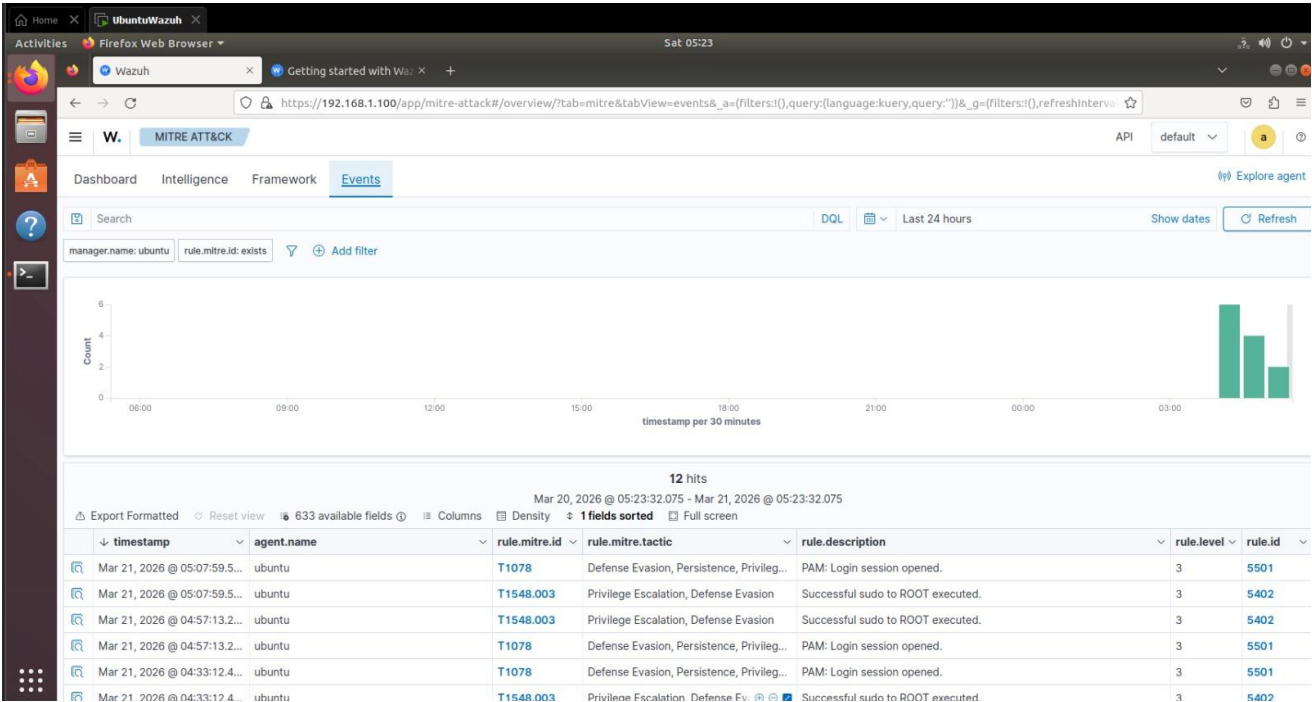
Mitre Attack Dashboard



Mitre Attack Framework

Tactics			
Privilege Escalation	11		
Defense Evasion	11		
Persistence	8		
Initial Access	7		
Credential Access	0		
Execution	0		
Impact	0		
Lateral Movement	0		
Exfiltration	0		
Discovery	0		
Techniques			
Filter techniques of selected tactic/s			
T1078 - Valid Accounts	7	T1548.003 - Sudo and Sudo Caching	4
T1556.003 - Pluggable Authentication M...	0	T1056.001 - Keylogging	0
T1171 - LLMNR/NBT-NS Poisoning and R...	0	T1539 - Steal Web Session Cookie	0
T1555.002 - Securityd Memory	0	T1522 - Cloud Instance Metadata API	0
T1003.004 - LSA Secrets	0	T1606.002 - SAML Tokens	0
T1003.007 - Proc Filesystem	0	T1555.005 - Password Managers	0
T1556.002 - Password Filter DLL	0	T1558.004 - AS-REP Roasting	0
T1552 - Unsecured Credentials	0	T1139 - Bash History	0
T1136 - Create Account	1	T1557 - Adversary-in-the-Middle	0
T1110.001 - Password Guessing	0	T1003 - OS Credential Dumping	0
T1003.002 - Security Account Manager	0	T1552.005 - Cloud Instance Metadata API	0
T1110.002 - Password Cracking	0	T1555.001 - Keychain	0
T1167 - Securityd Memory	0	T1214 - Credentials in Registry	0
T1040 - Network Sniffing	0	T1552.002 - Credentials in Registry	0
T1558 - Steal or Forge Kerberos Tickets	0	T1555 - Credentials from Password Stores	0
T1503 - Credentials from Web Browsers	0	T1556.007 - Hybrid Identity	0

Mitre Attack Events



Mitre Attack Intelligence

The screenshot displays the Wazuh MITRE ATT&CK Intelligence page. The top navigation bar includes 'Dashboard', 'Intelligence', 'Framework', and 'Events'. The 'Intelligence' tab is active. A search bar is present with filters for 'manager.name: ubuntu' and 'rule.mitre.id: exists'. A table lists 150 threat groups, with columns for ID, Name, and Description. The table shows three groups: G0018 (admin@338), G0130 (Ajax Security Team), and G0138 (Andariel).

ID	Name	Description
G0018	admin@338	admin@338 is a China-based cyber threat group. It has previously used newsworthy events as lures to deliver malware and has primarily targeted organizations involved in financial, economic, and trade policy, typically using publicly available RATs such as PoisonIvy, as well as some non-public backdoors. (Citation: FireEye admin@338)
G0130	Ajax Security Team	Ajax Security Team is a group that has been active since at least 2010 and believed to be operating out of Iran. By 2014 Ajax Security Team transitioned from website defacement operations to malware-based cyber espionage campaigns targeting the US defense industrial base and Iranian users of anti-censorship technologies. (Citation: FireEye Operation Saffron Rose 2013)
G0138	Andariel	Andariel is a North Korean state-sponsored threat group that has been active since at least 2009. Andariel has primarily focused its operations—which have included destructive attacks—against South Korean government agencies, military organizations, and a variety of domestic companies; they have also conducted cyber financial operations against ATMs, banks, and cryptocurrency exchanges. Andariel's notable activity includes Operation Black Mine, Operation GoldenAxe, and Campaign Rifle. (Citation: FSI Andariel Campaign Rifle July 2017) (Citation: IssueMakersLab Andariel GoldenAxe May 2017) (Citation: AhnLab Andariel Subgroup of Lazarus June 2018) (Citation: TrendMicro New Andariel Tactics July 2018) (Citation: CrowdStrike Silent Chollima Adversary September 2021) Andariel is considered a sub-set of Lazarus Group, and has been attributed to North Korea's Reconnaissance General Bureau. (Citation: Tencent North Korea Cyber Group September 2019)